

One-year follow-up showing effects of single intra-articular injection of hyaluronic acid (1,500-2,000 kDa) in symptomatic knee osteoarthritis.

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Clinical evidence on knee osteoarthritis suggests that intra-articular administration of hyaluronic acid may be useful in the management of patients with persistent pain. This study assesses the duration of effectiveness of a single intra-articular hyaluronic acid injection in a large population of patients with knee osteoarthritis. This retrospective post-marketing cohort study collected data from the ANTIAGE Registry (<http://www.antiagefbf.it/registro>), selecting patients of age ≥ 40 years, with symptomatic knee osteoarthritis (Kellgren-Lawrence grade I-III) of ≥ 12 months duration, and ≥ 12 months of follow-up. Patients had received a single intra-articular injection of high molecular weight hyaluronic acid (1,500-2,000 kDa) at baseline. WOMAC Osteoarthritis Index total scores measured using the LK 3.1 scale and 10 cm VAS pain scores were evaluated before IA Injection and at 6, 9, 10, 11 and 12 months. Blood cell counts, uricemia, erythrocyte sedimentation rates and levels of C-reactive protein were measured at baseline and 12 months. Time from initial treatment to second injection up to 12 months was recorded to assess event-free survival. Included patients ($n=187$) were 53.5% female and had a mean ($\pm$ SD) age at baseline of 62 (± 16.6) years and mean ($\pm$ SD) body mass index of 26.2 (± 2.5) kg/m². Mean ($\pm$ SD) WOMAC index total score and VAS pain scores were 60.9 (± 7.1) and 5.9 cm (± 1.8), respectively. There were statistically significant reductions compared to baseline in mean WOMAC index total score and VAS pain score at all time points (p less than 0.01 at 6 and 9 months; p less than 0.05 at 10, 11 and 12 months for both parameters). These results support the clinical effectiveness and safety of hyaluronic acid for up to 12 months for pain relief and function improvement in patients with knee osteoarthritis, confirming previous data on intra-articular administration of hyaluronic acid as chronic therapy in the management of knee osteoarthritis.

Evaluation of hyaluronic acid intra-articular injections in the treatment of primary and secondary osteoarthritis of the knee.

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Osteoarthritis is one of the most common locomotor conditions and the knee is the second most frequently affected articulation. One of therapeutic methods is viscosupplementation involving intra-articular injections of hyaluronic acid preparations. The aim of the study was to evaluate the clinical status and analyse selected biochemical blood parameters in patients with knee osteoarthritis after intra-articular injections of a hyaluronic acid preparation. Blood was taken from 113 patients, 60 patients with primary and 53 patients with secondary osteoarthritis, who were administered intra-articular injections of a hyaluronic acid preparation into the affected knee, according to the schedule established in the study protocol. Concentration of lipid hydroperoxides in blood plasma was determined, along with that of malondialdehyde in erythrocytes. Plasma concentrations of ceruloplasmin, TAC, albumin, total bilirubin, and uric acid were also determined. Clinical assessment was performed using a modified HHS scale and the WOMAC questionnaire. Intra-articular administration of a hyaluronic acid preparation significantly reduced pain in the knee and improved its functioning, regardless of the form of osteoarthritis. Not only clinical improvement was observed, but also beneficial changes in the blood antioxidant system. Comparable clinical improvement was observed in both forms of osteoarthritis of the knee. It was also beneficial changes in blood antioxidant system, especially in the group with secondary osteoarthritis.

TET enzymes, TDG and the dynamics of DNA demethylation.

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DNA methylation has a profound impact on genome stability, transcription and development. Although enzymes that catalyse DNA methylation have been well characterized, those that are involved in methyl group removal have remained elusive, until recently. The transformative discovery that ten-eleven translocation (TET) family enzymes can oxidize 5-methylcytosine has greatly advanced our understanding of DNA demethylation. 5-Hydroxymethylcytosine is a key nexus in demethylation that can either be passively depleted through DNA replication or actively reverted to cytosine through iterative oxidation and thymine DNA glycosylase (TDG)-mediated base excision repair. Methylation, oxidation and repair now offer a model for a complete cycle of dynamic cytosine modification, with mounting evidence for its significance in the biological processes known to involve active demethylation.

Tet proteins: on track towards DNA demethylation?

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Abstract Dynamic DNA methylation is a prerequisite for many developmental processes and maintenance of cellular integrity. In mammals however, mechanisms of active DNA demethylation have for long been elusive. The discovery of the ten-eleven translocation (Tet) family of enzymes that oxidize 5-methylcytosine (5mC) to 5-hydroxymethylcytosine (5hmC), 5-formylcytosine (5fC) or 5-carboxylcytosine (5caC) provided new means by which DNA methylation could actively be reversed. This review focuses on the possible mechanisms of DNA demethylation via Tet proteins and their metabolites 5hmC, 5fC and 5caC. Additionally, it discusses the roles of the three Tet protein family members Tet1, Tet2 and Tet3 as developmental regulators, probably in part independent of their enzymatic activity. By contrast, recent evidence suggests a function of 5hmC as an epigenetic mark on its own, going beyond the expectation of only acting as an intermediate in an active DNA demethylation pathway.

Biochemical reconstitution of TET1-TDG-BER-dependent active DNA demethylation reveals a highly coordinated mechanism.

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Cytosine methylation in CpG dinucleotides is an epigenetic DNA modification dynamically established and maintained by DNA methyltransferases and demethylases. Molecular mechanisms of active DNA demethylation began to surface only recently with the discovery of the 5-methylcytosine (5mC)-directed hydroxylase and base excision activities of ten-eleven translocation (TET) proteins and thymine DNA glycosylase (TDG). This implicated a pathway operating through oxidation of 5mC by TET proteins, which generates substrates for TDG-dependent base excision repair (BER) that then replaces 5mC with C. Yet, direct evidence for a productive coupling of TET with BER has never been presented. Here we show that TET1 and TDG physically interact to oxidize and excise 5mC, and proof by biochemical reconstitution that the TET-TDG-BER system is capable of productive DNA demethylation. We show that the mechanism

assures a sequential demethylation of symmetrically methylated CpGs, thereby avoiding DNA double-strand break formation but contributing to the mutability of methylated CpGs.

5-Methylcytosine DNA demethylation: more than losing a methyl group.

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Demethylation of 5-methylcytosine in DNA is integral to the maintenance of an intact epigenome. The balance between the presence or absence of 5-methylcytosine determines many physiological aspects of cell metabolism, with a turnover that can be measured in minutes to years. Biochemically, addition of the methyl group is shared among all living kingdoms and has been well characterized, whereas the removal or reversion of this mark seems diverse and much less understood. Here, we present a summary of how DNA demethylation can be initiated directly, utilizing the ten-eleven translocation (TET) family of proteins, activation-induced deaminase (AID), or other DNA modifying enzymes, or indirectly, via transcription, RNA metabolism, or DNA repair; how intermediates in those pathways are substrates of the DNA repair machinery; and how demethylation pathways are linked and possibly balanced, avoiding mutations.

TET3-facilitated differentiation of human umbilical cord mesenchymal stem cells into oligodendrocyte precursor cells for spinal cord injury recovery.

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Spinal cord injury (SCI) inflicts a severe burden on patients and lacks effective treatments. Owing to the poor regenerative capabilities of endogenous oligodendrocyte precursor cells (OPCs) following SCI, there is a growing interest in alternative sources, such as human umbilical cord mesenchymal stem cells (HUCMSCs). TET3 is a key DNA demethylase that plays an important role in neural differentiation, but its role in OPC formation is not well understood. This study aimed to explore the TET3-mediated one-step induction of HUCMSCs into OPCs. In vitro, HUCMSCs were induced into OPCs following TET3 overexpression. Changes of methylation and hydroxymethylation during differentiation were monitored, mechanisms involved in the TET3-driven HUCMSC differentiation into OPCs were identified by RNA sequencing. Methylation levels in NG2 and PDGFRA promoter region were detected using Bisulfite Polymerase Chain Reaction (BSP). In vivo, therapeutic effects of iOPCs were evaluated through a rat Allen's SCI model. The in vitro analysis confirmed that TET3 enhances HUCMSC differentiation into OPCs, validated by specific marker expression. The induced OPCs (iOPCs) exhibited methylation and hydroxymethylation patterns similar to native OPCs. BSP analysis demonstrated that TET3 overexpression significantly reduced CpG island methylation in the NG2 and PDGFRA promoter regions. RNA sequencing revealed that TET3 induces iOPCs to express a series of genes essential for OPC formation while inhibiting the signaling pathways that hinder OPC development. In a rat model of SCI, TET3-overexpressing HUCMSCs appear to have the potential to differentiate into iOPCs in vivo, suppressed secondary injury, and promoted functional recovery. The therapeutic effects of iOPCs on SCI were superior to those of standard mesenchymal stem cell treatments. Our study demonstrated that TET3-mediated demethylation reshapes the methylation patterns of HUCMSCs, enabling their efficient one-step conversion into OPCs and significantly reducing the time required for cell preparation. This approach offers a potential strategy for early intervention in SCI. In an SCI model, TET3-induced OPCs

contributed to spinal cord repair, providing novel insights into cell therapy strategies for SCI through the lens of methylation regulation.

Human HDAC6 senses valine abundancy to regulate DNA damage.

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As an essential branched amino acid, valine is pivotal for protein synthesis, neurological behaviour, haematopoiesis and leukaemia progression¹⁻³. However, the mechanism by which cellular valine abundancy is sensed for subsequent cellular functions remains undefined. Here we identify that human histone deacetylase 6 (HDAC6) serves as a valine sensor by directly binding valine through a primate-specific SE14 repeat domain. The nucleus and cytoplasm shuttling of human, but not mouse, HDAC6 is tightly controlled by the intracellular levels of valine. Valine deprivation leads to HDAC6 retention in the nucleus and induces DNA damage. Mechanistically, nuclear-localized HDAC6 binds and deacetylates ten-eleven translocation 2 (TET2) to initiate active DNA demethylation, which promotes DNA damage through thymine DNA glycosylase-driven excision. Dietary valine restriction inhibits tumour growth in xenograft and patient-derived xenograft models, and enhances the therapeutic efficacy of PARP inhibitors. Collectively, our study identifies human HDAC6 as a valine sensor that mediates active DNA demethylation and DNA damage in response to valine deprivation, and highlights the potential of dietary valine restriction for cancer treatment.

Hypermethylation of CDKN2A CpG island drives resistance to PRC2 inhibitors in SWI/SNF loss-of-function tumors.

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Polycomb repressive complex 2 (PRC2) catalyzes the writing of the tri-methylated histone H3 at Lys27 (H3K27me3) epigenetic marker and suppresses the expression of genes, including tumor suppressors. The function of the complex can be partially antagonized by the SWI/SNF chromatin-remodeling complex. Previous studies have suggested that PRC2 is important for the proliferation of tumors with SWI/SNF loss-of-function mutations. In the present study, we have developed an EED-directed allosteric inhibitor of PRC2 termed BR0063, which exhibits anti-proliferative properties in a subset of solid tumor cell lines harboring mutations of the SWI/SNF subunits, SMARCA4 or ARID1A. Tumor cells sensitive to BR0063 exhibited several distinct phenotypes, including cell senescence, which was mediated by the up-regulation of CDKN2A/p16. Further experiments revealed that the expression of p16 was suppressed in the BR0063-resistant cells via DNA hypermethylation in the CpG island (CGI) promoter region, rather than via PRC2 occupancy. The expression of TET1, which is required for DNA demethylation, was found to be inversely correlated with p16 CGI methylation, and this may serve as a biomarker for the prediction of resistance to PRC2 inhibitors in SWI/SNF LOF tumors.

Injured inflammatory environment overrides the TET2 shaped epigenetic landscape of pluripotent stem cell derived human neural stem cells.

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Spinal cord injury creates an inflammatory microenvironment that regulates the capacity of transplanted human Neural Stem Cells (hNSC) to migrate, differentiate, and repair injury. Despite similarities in gene expression and markers detected by immunostaining, hNSC populations exhibit heterogeneous therapeutic potential. This heterogeneity derives in part from the epigenetic landscape in the hNSC genome, specifically methylation (5mC) and hydroxymethylation (5hmC) state, which may affect the response of transplanted hNSC in the injury microenvironment and thereby modulate repair capacity. We demonstrate a significant up-regulation of methylcytosine dioxygenase 2 gene (TET2) expression in undifferentiated hNSC derived from human embryonic stem cells (hES-NSC), and report that this is associated with hES-NSC competence for differentiation marker expression. TET2 protein catalyzes active demethylation and TET2 upregulation could be a signature of pluripotent exit, while shaping the epigenetic landscape in hES-NSC. We determine that the inflammatory environment overrides epigenetic programming in vitro and in vivo by directly modulating TET2 expression levels in hES-NSC to change cell fate. We also report the effect of cell fate and microenvironment on differential methylation 5mC/5hmC balance. Understanding how the activity of epigenetic modifiers changes within the transplantation niche in vivo is crucial for assessment of hES-NSC behavior for potential clinical applications.